Lecture 28

Quantifying with sequencing

How to compute coverages?

What is quantification?

1. Correlate a **biological process** with DNA fragments.
2. Sequence the DNA fragments.
3. Connect the **abundance** of fragments to the **rate** of the biological process.

Typically named as: **SOMETHING**-Seq

Examples: **ChIP**-Seq, **RNA**-Seq, **RAD**-Seq

All work by co-opting an existing biological mechanism to produce DNA fragments.

What is a Functional Assay?

A **biological process** represents a **biological function**.

Thus the **abundance of fragments** ought to correlate with the **activity level of the function**.

There are ongoing debates on the range of validity of each approach.

We "force" the **biological function** to produce **DNA fragments** that we can sequence.

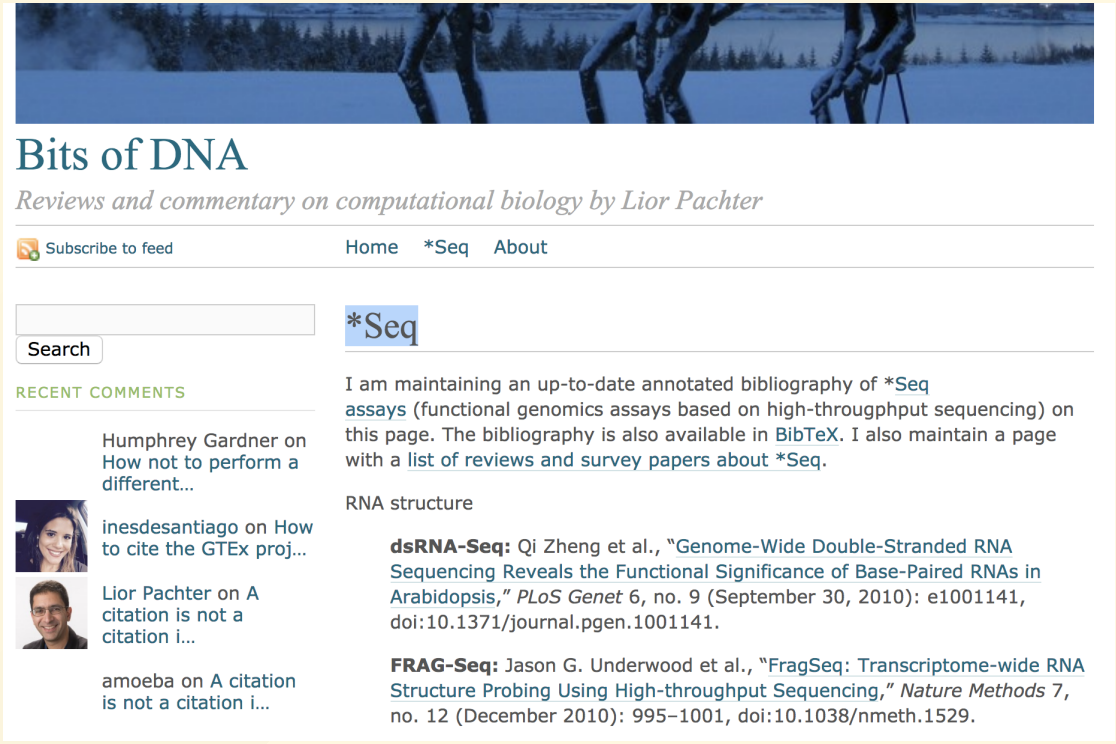
Common misconceptions

It is very common to hear someone misinterpret what a function assay does.

Example: *"ChIP-Seq measures the rate of protein binding to DNA."*

Reality is more complicated: *"ChIP-Seq measures how frequent DNA fragments are. Bound DNA locations produce more fragments than unbound locations"*

What types of Functional Assays exist?



Bits of DNA

Reviews and commentary on computational biology by Lior Pachter

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*Seq

I am maintaining an up-to-date annotated bibliography of *Seq assays (functional genomics assays based on high-througphput sequencing) on this page. The bibliography is also available in BibTeX. I also maintain a page with a list of reviews and survey papers about *Seq.

RNA structure

dsRNA-Seq: Qi Zheng et al., "Genome-Wide Double-Stranded RNA Sequencing Reveals the Functional Significance of Base-Paired RNAs in Arabidopsis," *PLoS Genet* 6, no. 9 (September 30, 2010): e1001141, doi:10.1371/journal.pgen.1001141.

FRAG-Seq: Jason G. Underwood et al., "FragSeq: Transcriptome-wide RNA Structure Probing Using High-throughput Sequencing," *Nature Methods* 7, no. 12 (December 2010): 995–1001, doi:10.1038/nmeth.1529.

Blog page: *Seq maintained by Prof. Lior Pachter.

RNA structure

- dsRNA-Seq, FRAG-Seq, SHAPE-Seq, DMS-Seq

Chromatin structure

- ATAC-Seq, FAIRE-Seq, ChIA-PET-Seq, Nucleo-Seq

Transcription

- RNA-Seq, GRO-Seq, 3-Seq, TIF-Seq

Translation

- Ribo-Seq, Frac-Seq, GTI-Seq

Will the Functional Sequencing take over the world?

It already did.

The way DNA "functions" is more important and essential to understanding life then the way DNA "looks".

How do I analyze *Seq data?

The essential requirement is to understand in what way does the data deviate from random DNA fragmentation.

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1. Understand what gets sequenced.
2. Understand what can and cannot be quantified.

Both can be suprisingly challenging get right - a typical mistake is to assume more than what a method does in reality.

For example for RNA-Seq

A cell has RNA of various types: `rRNA`, `mRNA`, `tRNA`, `snoRNA`, `miRNA`, etc.

- `mRNA` needs to be purified from total RNA.
- `RNA` needs to be reverse transcribed into DNA.
- Transcripts express at very different levels.
- Isoforms may be very similar?
- What detection level?
- Is the strand information retained?

Each step introduces its own biases and challenges.

The methods are similar

The analysis methods are typically very similar (perhaps with little twists).

- Align
- Quantify
- Compare

Interpreting the results is different. Now the specifics of the **origin** of the DNA matter A LOT.

Understanding a `*Seq` method means understaning the origin of the DNA.

Base coverage

Abundance based functional assays are all about coverages.

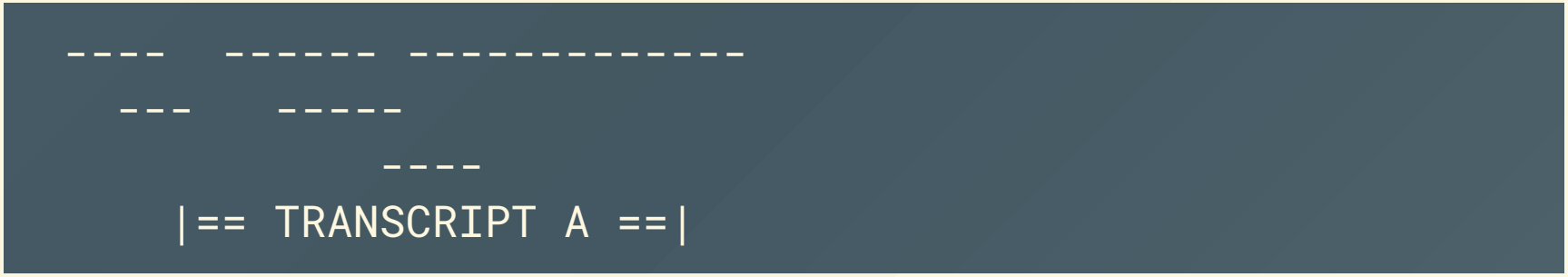
Coverage is the numer of reads that "cover" something.

Base coverage:

```
  ----   ------ -------
    -----   -----
               ----
01122112112222322211110
```

Coverage: how many sequencing reads cover a base.

Interval coverage



What is the coverage of transcript A? Is it

- `3`, `4` or `5`?

Depends on what "counting strategy" we use

`3` if all reads must be fully inside.
`4` if at least 50% of a read must be inside.
`5` if any overlap counts as a coverage.

*-Seq is all about coverages (1)

Perhaps you'll have to compare across different conditions:

1. Compute the coverage over **a thing** in condition **1** --> it produces a number: **100**
2. Compute the coverage over **the same thing** in condition **2** --> it produces another number: **200**

Do we have sufficient evidence that the two numbers represent a change?

*-Seq is all about coverages (2)

Compare across elements under the same condition:

1. Compute the coverage over **a thing** in condition **1** --> it produces a number: **100**
2. Compute the coverage over **another thing** in condition **1** --> it produces another number: **200**

Do we have sufficient evidence that the two numbers represent a change?

How to count coverages

Dealing with coverages is a universally useful skill applicable to many domains.

There are RNA-Seq specific approaches but it is important to understand how to do it in a generic way.

Start with the `megaton.sh` that we used for **Lecture 26**. This is why is good to have megaton scripts lying around.

```
http://data.biostarhandbook.com/variant/megaton.sh
bash megaton.sh
```

Coverages with `samtools`

Samtools already allows you to specify one or more ranges with the in a **chrom:start-end** format:

```
samtools view -c SRR1972739.bam AF086833:1-1000
```

Prints

```
141
```

You can combine the query with `SAM` flags as well:

```
samtools view -c -f 16 SRR1972739.bam AF086833:1-1000
54
```

`samtools` can take interval files

You can also store intervals in a BED file

```
AF086833        1       1000
AF086833        2000    3000
AF086833        5000    9000
```

Then

```
samtools view -c -L intervals.bed SRR1972739.bam
```

Prints:

```
953
```

`bedtools` the swiss army knife

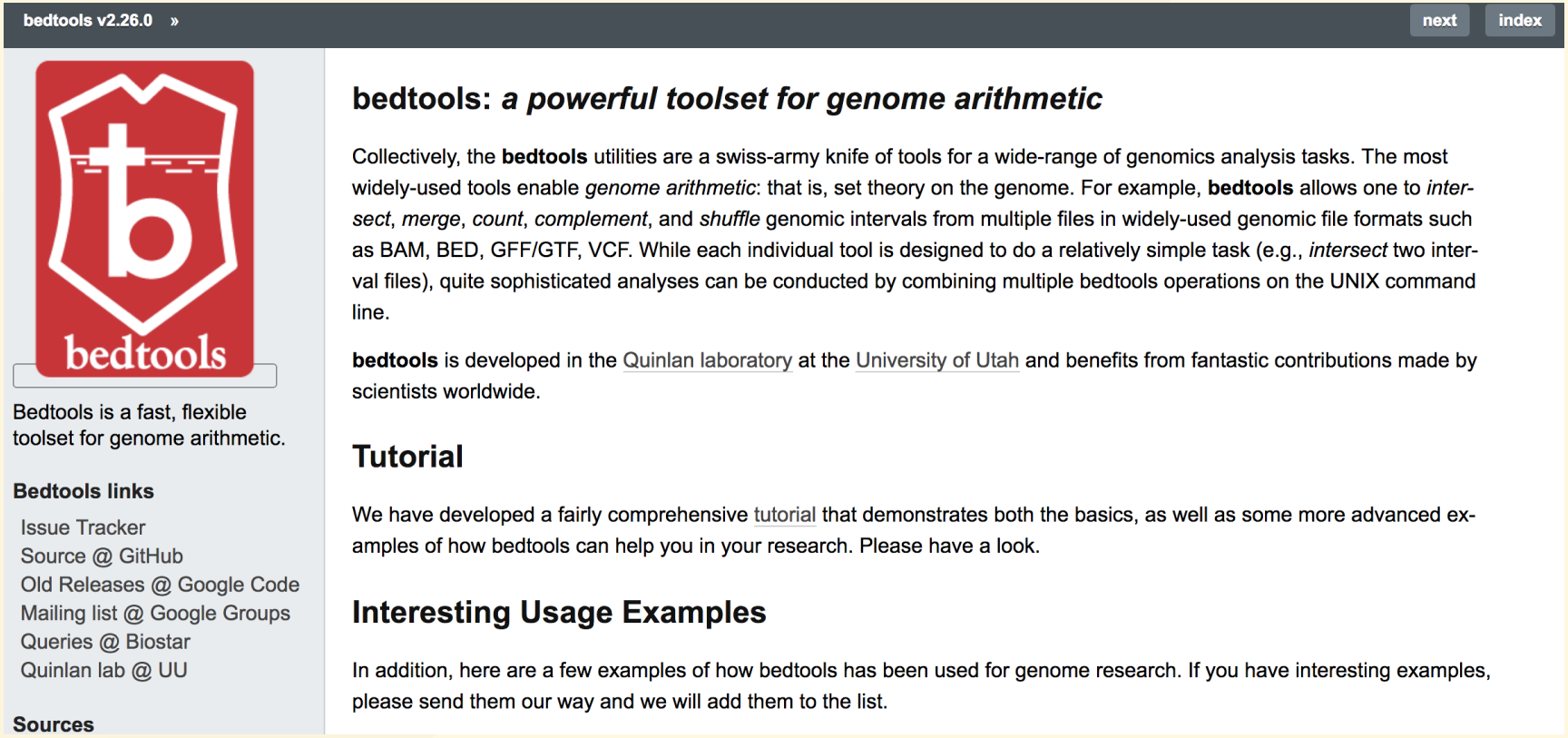
If you have to operate on annotations (interval files) of any kind `bedtools` might just be the most useful tool in your toolkit.

```
bedtools: flexible tools for genome arithmetic and DNA seq
usage:    bedtools <subcommand> [options]

The bedtools sub-commands include:

[ Genome arithmetic ]
    intersect     Find overlapping intervals in various wa
    window        Find overlapping intervals within a wind
    closest       Find the closest, potentially non-overla
    coverage      Compute the coverage over defined interv
    map           Apply a function to a column for each ov
    genomecov     Compute the coverage over an entire geno
    merge         Combine overlapping/nearby intervals int
```

Excellent documentation



Well documented with many examples.

`bedtools` essential utilites

A toolset that provides utilites to deal with intervals.

1. Create new intervals based on existing ones.
2. Extract or invert intervals.
3. Select or count overlapping intervals.
4. Extract sequences from a genome based on intervals.

`bedtools` undertstands both `BED`, `GFF` `VCF` and `BAM` files and makes use of them correctly.

Bam file coverage with `bedtools`

You can list multiple BAM files. I list only one here:

```
bedtools multicov -bams SRR1972739.bam
                        -bed db/AF086833.gff > counts.txt
```

Creates a new file with coverage values last.

```
cat counts.txt | cut -f 1,3,10 | head
AF086833        source  3106
AF086833        5'UTR   0
AF086833        gene    430
AF086833        mRNA    430
AF086833        regulatory      0
```

Will the coverage number match?

If you count the same way

```
cat counts.txt | cut -f 1,3,4,5,10 | head -1
AF086833        source  1       18959   3106
```

This does not match:

```
samtools view -c SRR1972739.bam AF086833:1-18959
3107
```

But this does (counting can be further cusomized):

```
samtools view -c -F 4 SRR1972739.bam AF086833:1-18959
3106
```